Decoupling Credibility from Conversion in Human and Virtual Influencer Marketing

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Abstract **Suzie Garcia (Wright)**

Decoupling Credibility from Conversion in Human and Virtual Influencer Marketing

This study examines the "Trust-Action Gap" in influencer marketing by comparing consumer reactions to human versus virtual influencers across Gen Z and Millennial cohorts. Drawing on the Theory of Planned Behavior and Source Credibility Theory, we conducted a between-subjects online experiment ($N=66$) to measure trust, authenticity, and purchase intention. Results revealed no significant generational differences in acceptance, challenging the assumption that "digital natives" are more receptive. However, a critical paradox emerged: while participants perceived human influencers as significantly more trustworthy and authentic ($p < .05$), this lack of trust in virtual influencers did not negatively impact Purchase Intention ($p > .05$). These findings suggest that consumers effectively decouple the "social agent" (emotional trust) from the "sales agent" (transactional utility), treating virtual influencers as sophisticated interactive displays. This study offers strategic insights for brands navigating the trade-off between emotional connection and transactional efficiency.

Keywords: virtual influencers, human-AI interaction (HAI), source credibility, digital authenticity, parasocial interaction, purchase intention, trust-action gap

Introduction **Suzie Garcia (Wright)**

For our group project, we are exploring the research question: **"Do younger generations (ages 18–28) or older generations (ages 29–44) trust recommendations from virtual influencers as much as human influencers?"** This question matters for two reasons.

In recent years, social media has transformed how people connect with brands and make buying decisions. Platforms like Instagram, TikTok, and YouTube have moved marketing away from traditional ads and toward influencer-driven content, where feeling a personal connection and seeing authenticity often makes all the difference. Within this world, virtual influencers, computer-generated characters with social media personas, have started to stand out. Unlike traditional influencers, these figures aren't real people, but they can still gain millions of followers, team up with big-name brands, and even take part in popular trends, blurring the line between what's real and what's digitally created.

From a research perspective, the first reason our question matters is that virtual influencers are changing how we think about advertising and how consumers engage with brands. For example, virtual personalities such as Lil Miquela, who has partnered with brands like Calvin Klein and Prada, blur the line between human and computer generated endorsers. These digital figures are not just novelty content, they actively engage followers, share lifestyle content, and participate in brand collaborations, making them a compelling case for studying authenticity and trust. The popularity of virtual influencers creates challenges and brings questions about the level of authenticity, relatability, and trustworthiness behind how consumers connect with brands and products.

Second, from a practical standpoint, brands are already investing heavily in virtual influencers, but it's not always clear how audiences respond to them compared to human influencers. Major brands are experimenting with virtual influencer campaigns to reach younger, digitally native audiences, yet there is limited evidence about how these investments impact engagement, purchase intention, or brand perception. Younger consumers are especially important to study because they are not only the most active users of social media but also trendsetters who shape the digital marketing landscape. Understanding whether they trust virtual influencers the same way they trust humans can help brands make smarter decisions about partnerships and campaigns. Moreover, examining generational differences can reveal whether openness to digital personas diminishes with age, providing actionable insights for targeting and engagement strategies.

Another factor to consider is familiarity with virtual influencers. Consumers who have prior exposure to virtual characters may perceive them as more trustworthy and relatable, while those unfamiliar with them may be more skeptical. This suggests that both demographic factors, such as age, and experiential factors, like prior exposure, could influence consumer responses, highlighting the complexity of trust in digital marketing contexts.

By exploring this topic, we hope to uncover insights that will help both researchers and marketers better understand how trust in influencers is changing in the digital age. Specifically, this study examines how age and familiarity with virtual influencers shape perceptions of trust, authenticity, purchase intention, and overall attitude toward the influencer.

Literature Review **Suzie Garcia (Wright)**

As virtual influencers become increasingly visible across social media platforms, marketers and researchers are exploring the question of whether computer-generated personas can genuinely replicate, or even enhance, the persuasive impact traditionally associated with human influencers. A growing body of research suggests that the effectiveness of virtual influencers hinges on how authenticity, emotional connection, trust, user characteristics, and the context of the products they promote interact. Understanding how these dimensions interact is critical for situating virtual influencers within broader theories of social influence and for anticipating how different audiences may respond to them.

One of the most foundational questions in this domain concerns whether people are capable of forming authentic emotional connections with virtual personas. The study *Virtual Influencers Versus Real Connections* (Mrad et al., 2024) directly examines this issue through the lens of parasocial relationship theory, which describes one-sided emotional bonds that individuals develop with celebrities, musicians, or human social media personalities. The research explores whether similar psychological processes occur when the "public figure" is a computer-generated entity rather than a real person. Findings suggest that audiences can develop parasocial relationships with virtual influencers, but the strength and quality of these connections depend heavily on perceived authenticity and emotional resonance. This introduces an early tension in the literature: virtual influencers are carefully designed to mimic human characteristics, yet their artificial nature makes it inherently difficult for them to replicate the emotional warmth or spontaneity that fuels many human–human parasocial bonds.

Questions about emotional connection also appear in studies outside the traditional influencer space. The research on *Consumer Responses to AI-Generated Charitable Giving Ads*

(Arango et al., 2023) illustrates how awareness of artificiality can dramatically alter emotional and behavioral reactions. Although the study does not focus specifically on virtual influencers, it provides important insights about how audiences respond when they discover that content designed to evoke empathy is generated by AI. Participants reported lower trust, reduced empathy, and decreased willingness to donate once they learned the images were artificial. These findings parallel concerns raised in the virtual influencer literature: the moment consumers recognize that emotional appeals have been constructed by AI, the sense of authenticity weakens. Interestingly, the study also notes contextual variation, AI-generated content can still be effective in cases where audiences do not expect or require human warmth. This nuance is important, as it suggests that virtual influencers may excel in certain marketing spaces while facing challenges in others.

Traditional research on human influencers provides an important point of comparison for understanding how virtual influencers differ. In the study *Social Media Influencers* (O'Donnell, 2022), O'Donnell shows that although the number of people who follow influencers has remained relatively stable, engagement among these followers has become deeper and more diverse. Users now follow influencers across several categories, from beauty and wellness to travel and gaming, and nearly 40 percent report purchasing products recommended by influencers or using influencer discount codes. These findings confirm the persuasive power of human influencers in driving consumer behavior and shaping brand relationships. When placed alongside virtual influencer research, this work helps clarify the strengths human influencers bring to marketing contexts, particularly their perceived personal experience, relatability, and authenticity, qualities that virtual influencers must compensate for through careful design and strategic branding.

Comparative research examining human and virtual influencers provides further clarity on how consumers interpret and evaluate these different types of endorsers. In *Close Encounters of the AI Kind* (Thomas & Fowler, 2021), Thomas and Fowler analyze how AI-generated influencers shape brand perceptions and consumer attitudes. They find that virtual influencers can successfully generate positive brand effects similar to those achieved by human celebrities, demonstrating that artificial personas can function effectively in promotional contexts. However, the study also uncovers an important discrepancy: AI influencers are perceived as highly interchangeable. If one virtual influencer makes an offensive or inappropriate statement, consumers tend to generalize the misconduct to all AI influencers, amplifying reputational risk for brands. In contrast, when human influencers behave poorly, the negative impact is often attributed to the individual rather than the category. This finding has significant strategic implications, as it suggests that brands may benefit from substituting an AI influencer with a human celebrity in the event of a scandal, an option that restores trust more effectively than simply replacing one virtual influencer with another. This generalizability of wrongdoing underscores a key limitation of virtual influencers: their lack of individuality in the eyes of consumers.

Another important aspect of how consumers interpret influencers is disclosure, specifically, how telling audiences that an influencer is not human affects their perceptions of credibility and trust. The article *Are They Humans or Robots?* (Wiley, 2023) contributes to this discussion by drawing on anthropomorphism theory and expectation disconfirmation theory. When audiences learn that an influencer is virtual, perceived human-likeness decreases sharply, leading to lower assessments of credibility and reduced trust in the associated brand. Notably, the study documents cross-cultural differences: U.S. consumers show strong negative reactions to

disclosure, whereas Brazilian consumers respond more moderately. These cultural distinctions hint at broader global variations in attitudes toward AI and digital identity, suggesting that brands using virtual influencers must consider both local norms and expectations regarding authenticity and human-like behavior.

While authenticity remains a central concern in this literature, the effectiveness of virtual influencers also depends on what they promote and how consumers evaluate the "fit" between influencer and product. A comparative study (Belanche et al., 2024) examines the interaction between influencer type (human versus virtual) and product type (hedonic versus utilitarian). The results show that virtual influencers do not inherently diminish purchase intention; instead, their effectiveness is mediated by perceived usefulness and identification. Virtual influencers were seen as more appropriate for utilitarian products, where competence and information are valued, whereas human influencers generated stronger identification in hedonic contexts, where emotional resonance is more important. This pattern reinforces the idea that virtual influencers may thrive in settings where technical accuracy, consistency, or efficiency matter more than human warmth or spontaneity. Conversely, when the product requires emotional engagement or experiential appeal, human influencers often retain an advantage.

Other recent studies also highlight the importance of user-specific factors that shape how consumers respond to virtual or AI-generated influencers. Lee and Liu argue that trust in AI-driven agents is not simply the result of the "machine heuristic," the assumption that machines are objective, rational, and free from human bias. Rather, trust is strongly influenced by consumers' AI literacy and familiarity (Lee & Liu, 2025). Individuals with greater experience and knowledge of AI technologies show higher levels of comfort and trust in AI-generated recommendations. This is highly relevant for virtual influencer research, as it

implies that younger generations or technologically sophisticated users may be more receptive to VIs, while those less familiar with AI may react with skepticism simply due to unfamiliarity.

Building on the idea that trust alone does not fully capture consumers' reactions to virtual influencers, Dondapati and Dehury expand the analytical framework by examining outcomes such as engagement, avoidance, and identification (Dondapati and Dehury, 2024). Their study shows that variables like homophily, authenticity, and parasocial relationships mediate the effects of influencer type. More importantly, authenticity acts as a crucial moderator. When users perceive a virtual influencer as authentic, regardless of their artificial nature, they are more likely to engage with and identify with the influencer. But when authenticity is lacking, virtual influencers elicit avoidance rather than persuasion. This highlights a consistent finding across studies: the effectiveness of virtual influencers relies less on their human status and more on their ability to convincingly mimic, or strategically convey, the cues of authentic human presence.

Taken together, the literature highlights that virtual influencers are capable of generating persuasive effects comparable to human influencers, but their effectiveness depends on multiple psychological, contextual, and demographic factors. Emotional connection and perceived authenticity consistently emerge as key drivers of trust, engagement, and identification with influencers, while product type and consumer characteristics further shape responses (Belanche et al., 2024; Mrad et al., 2024; Thomas & Fowler, 2021; Wiley, 2023). Age appears particularly important: younger consumers, especially Gen Z, show greater familiarity with AI-generated content and are generally more receptive to virtual influencers, whereas older adults tend to be more skeptical (Lee & Liu, 2025). In addition, human influencers are often perceived as more relatable, trustworthy, and capable of generating emotional resonance, particularly in hedonic contexts.

Based on this synthesis, the following hypotheses are proposed:

- **H1:** Gen Z consumers will show greater acceptance of products promoted by influencers than older adults.
- **H2:** Consumers will be more willing to accept products recommended by human influencers than by virtual influencers.
- **H3:** Gen Z consumers will have more positive attitudes and higher trust toward influencers than older adults.
- **H4:** Consumers will have more positive attitudes and higher trust toward human influencers than toward virtual influencers.

Methods

Bella Kang

Research Design and Rationale

We conducted a between-subjects online experiment to compare how Influencer Type (human influencer vs. virtual influencer) and Generational Cohort (Gen Z vs. Millennials) shape consumer reactions to skincare endorsement content.

Virtual influencers have rapidly entered commercial advertising spaces such as fashion and technology,yet empirical evidence on whether consumers trust them in high-involvement and credibility-sensitive domains like skincare remains limited. Because skincare purchases often rely on perceived authenticity and interpersonal credibility, understanding whether CGI-based endorsements can evoke similar responses to human endorsements offers both theoretical and industry value.

To test this, we held message content constant across influencer conditions and manipulated only the identity of the presenter. This allowed us to isolate whether identity alone could shift trust, perception, and behavioral intention. We additionally compared Generational Cohorts, given that Gen Z is generally more open to AI-mediated experiences and synthetic media, whereas Millennials tend to value authenticity cues and transparency in influencer communications. Generational contrast therefore offered a useful lens to examine digital socialization differences as a moderator of virtual influencer persuasion.

Participants

Participants were recruited using snowball sampling via Instagram posts, LinkedIn resharing, and university mailing lists, hosted on Qualtrics. Snowball sampling is commonly used in influencer and digital consumer research because it efficiently reaches naturally networked populations where social media usage is already embedded. Eligibility criteria required participants to be U.S. residents aged 18–44. To retain high data quality, responses failing attention or manipulation checks were removed, following standard online survey rigor. The final sample was N = 66, with a predominantly female demographic, which accurately reflects real skincare consumption trends. This enhances ecological validity rather than requiring artificial gender balancing.

Table 1

Descriptive Statistics for Participant Characteristics and Measurement Variables

Variable	Mean / Percentage (%)	SD
Participant Characteristics (*N* = 66)		
Age Group (18–28)	71.20	—
Gender (Female)	74.20	—
Social Media Usage (Daily)	95.50	—
Measurement Variables (*N* = 66)		
Product Recommendation	2.25	0.77
Purchase Intention	2.64	0.97
Positive Feeling (Attitude)	2.71	0.87
Trust / Authenticity	2.44	0.76

Note. The Measurement Variables were assessed on an unspecified scale (range 2.25 to 2.71).

Stimuli

All participants viewed a 15-second YouTube Shorts–style skincare endorsement video. Two visually parallel versions were produced under identical filming conditions:

- **Human Influencer:** Real woman presenting the serum.
- **Virtual Influencer:** Hyper-realistic CGI model styled in commercial VI aesthetic.

All script, tone, product placement, background, lighting, framing, and pacing were held constant, ensuring the only manipulated variable was influencer identity. The format reflected contemporary short-form advertising commonly encountered on TikTok, Instagram Reels, and YouTube Shorts, increasing ecological realism rather than presenting unfamiliar media structures. This design minimized novelty confounding and allowed differences to be attributed to influencer type alone.

Procedure

The experiment followed a standardized flow through Qualtrics. The average survey duration was approximately 8 minutes. All responses were recorded anonymously, and no personally identifiable information (PII) was retained. Finally, participants were thanked for their time, ensuring that the data collection process adhered to standard ethical guidelines for online research.

I. **Screening:** Participants self-reported age and were assigned to Gen Z (18–28) or Millennial (29–44) categories.
    - *Rationale:* Generational cohorts differ in AI trust and persuasion processing patterns.

II. **Random Assignment:** Participants were automatically assigned to Human vs. Virtual influencer conditions.
    - *Rationale:* Randomization reduces allocation bias and increases causal inference reliability.

III. **Exposure:** Participants viewed one embedded video (no skip option).
    - *Rationale:* Immediate single-exposure mimics real social media ad experience.

IV. **Measurement:** Participants completed self-report items on influencer evaluation, trust, and purchase intentions.
    - *Rationale:* Post-exposure reporting captures intuitive persuasion response before cognitive reconstruction.

V. **Attention/Manipulation Check:** Included one attention item and one influencer-recognition item.
    - *Rationale:* Removes satisficing behavior and maintains internal validity.

VI. **Debrief + Familiarity Item:** Participants reported recognition/familiarity with virtual influencers.
    - *Rationale:* Familiarity is known to increase openness to synthetic agents.

Measures

Measures All constructs used 5-point Likert scales (1 = strongly disagree → 5 = strongly agree). Items were adapted from validated persuasion and influencer credibility research.

Table 2

Results of Independent Samples t-Tests Comparing Mean Scores Across Influencer Type and Generational Cohort

Variable	Group 1 M	Group 2 M	t	p
Panel A: Influencer Type	Human	Virtual		
Product Recommendation	2.45	2.05	2.18	.033
Purchase Intention	2.76	2.52	0.99	.324
Positive Feeling (Attitude)	3.03	2.39	3.21	.002
Trust / Authenticity	2.68	2.20	2.67	.010
Panel B: Generational Cohort	Gen Z	Millennial		
Product Recommendation	2.33	2.07	1.22	0.226
Purchase Intention	2.69	2.52	0.66	0.513
Positive Feeling (Attitude)	2.82	2.42	1.73	0.089
Trust / Authenticity	2.49	2.34	0.73	0.470

Note. M = Mean. Independent samples t-tests were conducted to compare mean scores for four dependent variables across two groupings: Influencer Type (Human vs. Virtual) and Generational Cohort (Gen Z vs. Millennial). * $p < .05$. | ** $p < .01$.

Data Analysis

Data processing and statistical analyses were performed using PSPP. Prior to hypothesis testing, the data were screened for accuracy and missing values. Responses failing the attention and manipulation checks were excluded to ensure data quality.

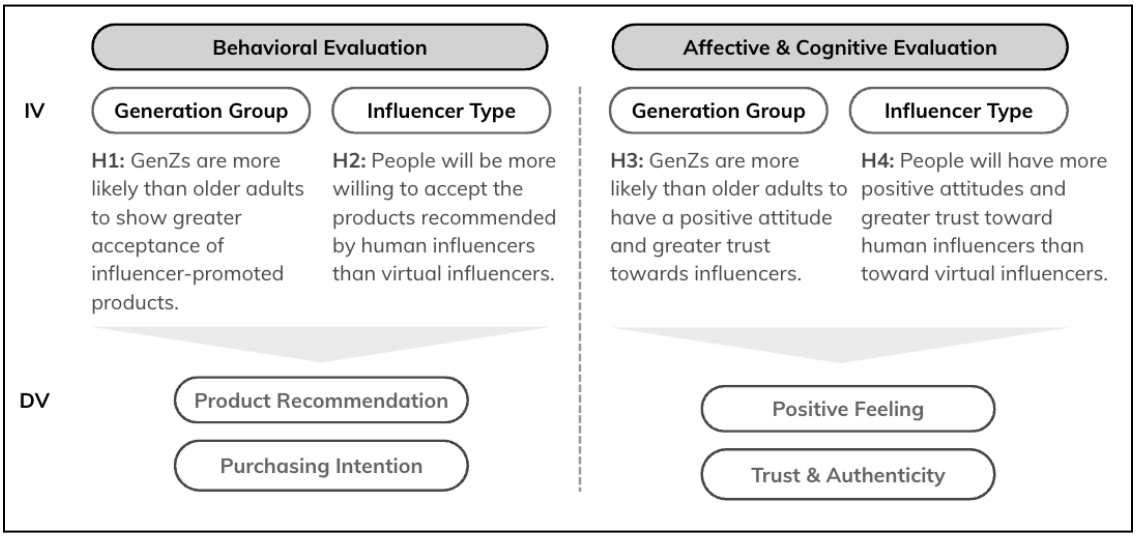
First, descriptive statistics (means, standard deviations, and frequencies) were generated to summarize participant demographics and check the distribution of dependent variables. Second, internal consistency reliability was assessed for all multi-item scales using Cronbach's α coefficients to ensure measurement validity.

To test the main hypotheses, independent samples *t*-tests were conducted. This parametric test was selected to compare mean differences in consumer responses across two categorical independent variables: (1) Influencer Type (Human vs. Virtual) and (2) Generational

Cohort (Gen Z vs. Millennials). Statistical significance was determined based on an alpha level of $\alpha = .05$. Additionally, exploratory analyses were performed to observe potential interaction trends between generation and influencer type.

Results **Annie Yang**

Image N.



The first hypothesis posits that *GenZs are more likely than older adults to show greater acceptance of influencer-promoted products*. We tested this hypothesis by examining how customers in class differed between the 18–28 age group and the 29–44 age group. We tested the first hypothesis using two dependent variables – product recommendation and purchase intention – both of which relate to **Behavioral Evaluation** within the CAB framework.

DV1: Product Recommendation

Table 2-1 reports the one of the results of independent samples t-test to examine this hypothesis for. Results indicate that participants in the 18–28 age group reported slightly higher product-recommendation attitudes (M = 2.33, n = 47) than those in the 29–44 age group (M = 2.07, n = 19). However, this difference **was not statistically significant**, t = 1.22, p = .226. This indicates that although younger participants appeared somewhat more favorable toward influencer-promoted products, the observed difference could be attributed to random variation.

Table 2-1. Independent Samples T-tests on acceptance of product recommendation from influencers between the 18–28 age group and the 29–44 age group

Independent Samples Test

		Levene's Test for Equality of Variances		T-Test for Equality of Means						
									95% Confidence Interval of the Difference	
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	Lower	Upper
product_recommend	Equal variances assumed	.28	.599	1.22	64.00	.226	.26	.21	-.16	.67
	Equal variances not assumed			1.20	32.33	.237	.26	.21	-.18	.69

Overall, the results do not provide sufficient evidence to support the first hypothesis that age group meaningfully affects behaviors toward influencer-promoted products.

DV2: Purchase Intension

Table 2-2 reports the one of the results of independent samples t-test to examine this hypothesis for. Results indicate that participants in the 18–28 age group reported slightly higher purchase intention (M = 2.69, n = 47) than those in the 29–44 age group (M = 2.52, n = 19). However, **this difference was not statistically significant**, t(64) = 0.66, p = .513. Although younger participants appeared marginally more willing to purchase influencer-promoted products, the results do not provide evidence to support the hypothesis that Gen Z is more likely to make such purchases compared to older adults.

Table 2-2. Independent Samples T-tests on the willingness of purchase intention from influencers between the 18–28 age group and the 29–44 age group

Independent Samples Test

		Levene's Test for Equality of Variances		T-Test for Equality of Means						
									95% Confidence Interval of the Difference	
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	Lower	Upper
mean_purchase_inte	Equal variances assumed	18.34	.000	.66	64.00	.513	.17	.26	-.35	.70
	Equal variances not assumed			.54	23.50	.596	.17	.32	-.49	.84

The second hypothesis posits that *People will be more willing to accept the products recommended by human influencers than virtual influencers*. We tested this hypothesis by examining how customers in class differed between the human influencer group and the virtual influencer group. We tested the second hypothesis also using two dependent variables – product recommendation and purchase intention – part of the same behavioral framework like the first hypothesis.

DV1: Product Recommendation

Table 3-1 reports the one of the results of independent samples t-test to examine this hypothesis for. Results indicate that participants in the human influencer condition reported higher product-recommendation ratings (M = 2.45, n = 33) than those in the virtual influencer condition (M = 2.05, n = 33). **This difference was statistically significant**, t(64) = 2.18, p = .033. This data provide sufficient evidence to conclude that participants were more receptive to product recommendations from human influencers than from virtual influencers. In short, this result supports the hypothesis that people tend to hold more willingness to accept product recommendations delivered by human influencers.

Table 3-1. Independent Samples T-tests on acceptance of product recommendation from influencers between the human influencer group and the virtual influencer group

Independent Samples Test

		Levene's Test for Equality of Variances		T-Test for Equality of Means						
									95% Confidence Interval of the Difference	
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	Lower	Upper
product_recommend:	Equal variances assumed	.89	.349	2.18	64.00	.033	.40	.19	.03	.77
	Equal variances not assumed			2.18	63.61	.033	.40	.19	.03	.77

DV2: Purchase Intention

Table 3-2 reports the one of the results of independent samples t-test to examine this hypothesis for. Results indicate that participants in the human influencer condition reported slightly higher purchase intentions (M = 2.76, n = 33) than those in the virtual influencer condition (M = 2.52, n = 33). However, **this difference was not statistically significant**, t(64) = 0.99, p = .324. Since the p-value exceeds the conventional .05 threshold, the data do not provide sufficient evidence to conclude that participants are more willing to purchase products recommended by human influencers than by virtual influencers. Thus, this result does not support the hypothesis that human influencers elicit more willingness to accept the products than virtual influencers.

Table 3-2. Independent Samples T-tests on the willingness of purchase intention from influencers between the human influencer group and the virtual influencer group

Independent Samples Test

		Levene's Test for Equality of Variances		T-Test for Equality of Means						
									95% Confidence Interval of the Difference	
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	Lower	Upper
mean_purchase_inte	Equal variances assumed	.88	.352	.99	64.00	.324	.24	.24	-.24	.71
	Equal variances not assumed			.99	63.56	.324	.24	.24	-.24	.71

In conclusion, since only one of the dependent variables supports the second hypothesis, this hypothesis is partially supported.

Moving beyond behavioral measures, the following analysis result explores the hypothesis within the domains of affective and cognitive evaluation.

The third hypothesis posits that *GenZs are more likely than older adults to have a positive attitude and greater trust towards influencers.* We tested this hypothesis by examining how customers in class differed between the 18–28 age group and the 29–44 age group. We tested the first hypothesis using two dependent variables – positive feeling and trust authenticity – both of which relate to **Affective & Cognitive Evaluation** within the CAB framework.

DV1: Positive Feeling

Table 4-1 reports the one of the results of independent samples t-test to examine this hypothesis for. Results indicate that participants in the 18–28 age group reported slightly more positive feelings toward influencers (M = 2.82, n = 47) than those in the 29–44 age group (M = 2.42, n = 19). However, **this difference was not statistically significant,** t(64) = 1.73, p = .089. Because the p-value exceeds the conventional .05 threshold, the data do not provide sufficient evidence to conclude that younger adults hold more positive attitudes than older adults. Thus, this result does not support hypothesis 3 for the positive-feeling variable.

Table 4-1. Independent Samples T-tests on the extent of positive feeling from influencers between the 18–28 age group and the 29–44 age group

Independent Samples Test

		Levene's Test for Equality of Variances		T-Test for Equality of Means						
									95% Confidence Interval of the Difference	
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	Lower	Upper
positive_feeling	Equal variances assumed	1.47	.229	1.73	64.00	.089	.40	.23	-.06	.87
	Equal variances not assumed			1.64	30.01	.112	.40	.25	-.10	.91

DV2: Trust & Authenticity

Table 4-2 reports the one of the results of independent samples t-test to examine this hypothesis for. Results indicate that participants in the 18–28 age group reported slightly higher trust and authenticity perceptions toward influencers (M = 2.49, n = 47) than those in the 29–44 age group (M = 2.34, n = 19). However, **this difference was not statistically significant**, t(64) = 0.73, p = .47. Because the p-value exceeds the conventional .05 threshold, the data do not provide sufficient evidence to conclude that younger adults perceive influencers as more trustworthy or authentic than older adults.

Table 4-2. Independent Samples T-tests on the extent of trust & authenticity feeling towards influencers between the 18–28 age group and the 29–44 age group

Independent Samples Test

		Levene's Test for Equality of Variances		T-Test for Equality of Means						
									95% Confidence Interval of the Difference	
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	Lower	Upper
trust_avg	Equal variances assumed	.77	.382	.73	64.00	.468	.15	.21	-.26	.57
	Equal variances not assumed			.66	27.21	.518	.15	.23	-.32	.62

Therefore, these results do not support the hypothesis that GenZs are more likely than older adults to have a positive attitude and greater trust towards influencers.

The final hypothesis argues that *People will have more positive attitudes and greater trust toward human influencers than toward virtual influencers.* We tested this hypothesis by examining how customers in class differed between the human influencer group and the virtual influencer group. Same as the previous hypothesis, we tested this hypothesis using two dependent variables – positive feeling and trust authenticity.

DV1: Positive Feeling

Table 5-1 reports the one of the results of independent samples t-test to examine this hypothesis for. Results indicate that participants in the human influencer condition reported significantly more positive feelings (M = 3.03, n = 33) than those in the virtual influencer condition (M = 2.39, n = 33). **This difference was statistically significant**, t(64) = 3.21, p = .002. Because the p-value is well below the conventional .05 threshold, the data provide strong evidence that participants experience more favorable attitudes when the influencer is human rather than virtual.

Table 5-1. Independent Samples T-tests on the extent of positive feeling from influencers between the human influencer group and the virtual influencer group

Independent Samples Test

		Levene's Test for Equality of Variances		T-Test for Equality of Means						
									95% Confidence Interval of the Difference	
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	Lower	Upper
positive_feeling	Equal variances assumed	4.28	.043	3.21	64.00	.002	.64	.20	.24	1.05
	Equal variances not assumed			3.21	60.55	.002	.64	.20	.24	1.05

DV2: Trust & Authenticity

Table 5-2 reports the one of the results of independent samples t-test to examine this hypothesis for. Results indicate that participants in the human influencer condition reported higher perceptions of trust and authenticity (M = 2.68, n = 33) than those in the virtual influencer condition (M = 2.20, n = 33). **This difference was statistically significant**, t(64) = 2.67, p = .010. Because the p-value falls below the conventional .05 threshold, the data provide clear evidence that participants viewed human influencers as more trustworthy and authentic than virtual influencers.

Table 5-2. Independent Samples T-tests on the extent of trust & authenticity feeling from influencers between the human influencer group and the virtual influencer group

Independent Samples Test

		Levene's Test for Equality of Variances		T-Test for Equality of Means						
									95% Confidence Interval of the Difference	
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	Lower	Upper
trust_avg	Equal variances assumed	.99	.323	2.67	64.00	.010	.48	.18	.12	.83
	Equal variances not assumed			2.67	57.76	.010	.48	.18	.12	.83

Thus, the above results support the final hypothesis that human influencers elicit more positive attitudes and greater trust perceptions than virtual influencers.

Discussion (Part 1) **Tanisha Acharya**

Thus to summarize the results, there is no difference between the age groups when it comes to acceptance of influencer promoted products, positive sentiment or trust in influencers. However, the type of influencer emerged as a significant determinant of consumer response. Overall people are much more willing to accept product recommendations from human influencers over virtual ones. People also feel more positively and are more trusting of human influencers over virtual ones. However, it must be noted that none of attitudinal disparities translate into behavioral outcomes –people are not more willing to buy from a human influencer than a virtual one.

These findings suggest distinct strategic applications for marketers. If the campaign objective prioritizes brand affinity, reputation management, or the establishment of trust, human influencers should be prioritized. The data indicates that genuineness and authenticity remain distinctly human traits in the eyes of the consumer. Human influencers possess a unique capacity to elicit deep emotional resonance and foster parasocial bonds, whereas, virtual influencers may risk alienating audiences where emotional connection is paramount.

Conversely, if the objective is strictly sales conversion, virtual influencers represent a viable strategy. The dissociation between affect/trust and behavioral intent implies that high levels of "warmth" or trust are not the sole drivers of purchase decisions. Thus brands could potentially use virtual influencers to drive sales. Additionally, from a risk management perspective, virtual influencers provide a distinct advantage over their human counterparts. Human influencers are susceptible to behavioral misconduct or public controversy – potentially leading to public scrutiny or 'cancellation' – which can cause reputational damage to the

associated brand. Whereas, virtual influencers allow for absolute control over messaging and conduct. Thus, virtual influencers also present a strategic advantage by ensuring consistent brand safety and mitigating the risk of negative publicity.

This study also challenges the categorization of influencer marketing as a niche "youth" phenomenon. The absence of significant differences in receptiveness between the older (29–44) and younger (18–28) cohorts suggests that marketers can broaden targeting parameters to include older millennials without fearing a decline in campaign effectiveness.

However, we must acknowledge several limitations of this study. First, the sample (N = 66) exhibited a demographic skew, with a predominance of younger participants (n = 47 aged 18–28 vs. n = 19 aged 29–44) and female respondents (n = 49). Second, the specific stimuli used—a skincare product presented via a "Get Ready With Me" (GRWM) video format—may limit the generalizability of the results to other product categories or content styles.

Future research should examine the interaction effects between independent variables, such as whether younger generations exhibit higher acceptance of virtual influencers compared to older cohorts. Subsequent studies should also diversify the product stimuli to include gender-neutral items (e.g., health supplements, energy drinks) or high-involvement luxury goods (e.g., automobiles) to assess external validity.

Finally, experimental designs should manipulate specific influencer attributes (e.g., ethnicity, age, degree of anthropomorphism) and content formats (e.g., direct product reviews and testimonials versus lifestyle integration) to figure out the specific drivers of consumer response.

Discussion (Part 2) **Bella Kang**

Interpretation of Findings

Our data suggests that while the *affective attitude*, which encompasses warmth and emotional connection, clearly favors humans, the *cognitive attitude*, which involves the assessment of product utility and information quality, remains stable across both influencer types. This finding presents a compelling contradiction that warrants a deeper examination through the lens of the Theory of Planned Behavior (Ajzen, 1991). According to this theory, behavioral intention is determined by a combination of attitude toward the behavior, subjective norms, and perceived behavioral control.

To further explain why participants exhibited robust purchase intentions toward virtual influencers despite lower trust, we refer to the Computers as Social Actors paradigm (Reeves & Nass, 1996). This theory suggests that individuals tend to apply social rules and expectations to computers and artificial agents, even when they consciously know these agents are not human. Our findings indicate a "mindless" social response in the specific context of purchasing. Participants processed the virtual influencer's endorsement as a valid social cue or recommendation without necessarily engaging in the "mindful" evaluation of the influencer's humanity. This aligns with the concept of source credibility in a hyper-mediated environment. Traditionally, credibility is composed of expertise and trustworthiness. In our study, the visual quality of the virtual influencer, or its expertise in presentation, may have compensated for the lack of human soul or trustworthiness. This implies that in the fast-paced environment of short-form video platforms like TikTok or YouTube Shorts, *visual authority* may effectively override the need for *ontological authenticity*.

Theoretical Implications

Our results complicate and extend current debates on digital authenticity. Belk (2021) argues that in the digital age, authenticity is increasingly fragmented, yet consumers still crave a core self behind the avatar. The significantly higher positive feelings toward human influencers in our study support Belk's assertion; consumers still value the perceived humanity as a gold standard for emotional connection. However, the fact that virtual influencers were accepted as valid marketing channels aligns with research on Human AI Interaction (Sundar & Kim, 2021). They contend that familiarity and uncertainty reduction are key determinants in accepting AI agents. Although trusted less, the virtual influencers in our stimuli were visually familiar enough to fit the participants' influencer schema. This suggests that consumers may be developing a specialized mental model for virtual agents where truth is not a prerequisite for influence, allowing them to process marketing messages without the suspension of disbelief required for human-like connection.

Moreover, our findings can be interpreted through the lens of the Proteus Effect (Yee & Bailenson, 2007). Studies from the Virtual Human Interaction Lab have demonstrated that embodied avatars significantly influence physical behavior through aspirational mirroring. In the context of skincare marketing, the virtual influencer serves as a hyper-idealized mirror. Even though consumers know the influencer's skin is computer generated, this visual perfection functions as a potent motivational prime. The virtual influencer does not just sell a serum; it sells the *possibility* of digital-grade perfection. This explains why Purchase Intention remained high despite lower trust scores. The virtual influencer successfully activated the consumer's desire for self-improvement through visual aspiration, a mechanism distinct from interpersonal trust.

Ethical and Societal Implications

The rise of virtual influencers introduces profound ethical questions regarding beauty standards and the future of creative labor. The virtual influencers used in our stimuli were designed with symmetrical features and flawless skin, an unattainable standard for biological humans. If virtual influencers become the dominant force in beauty marketing, they risk exacerbating body dysmorphia and unrealistic aesthetic expectations among younger generations. Unlike human influencers, who increasingly embrace body positivity and unfiltered content, virtual influencers are, by definition, engineered perfection. Marketers have a moral imperative to consider whether replacing human diversity with algorithmic perfection aligns with their corporate social responsibility goals. Additionally, the danger lies in the *illusion of empathy*; without biological consequences, virtual agents may promote unrealistic standards with algorithmic efficiency but without ethical restraint (Stanford HAI, 2023). This necessitates a clear distinction in their deployment and potentially stricter regulatory oversight compared to human influencers who bear the physical consequences of their own advice.

Conclusion: The Decoupling of Affect and Cognition

In conclusion, our study suggests that in the context of influencer marketing, consumers are effectively decoupling the "social agent" from the "sales agent." According to the Theory of Planned Behavior, behavioral intention is driven by attitude, subjective norms, and perceived behavioral control. Our data confirms that while the *affective attitude* (warmth, emotional connection) favors humans, the *cognitive attitude* (assessment of product utility and information quality) remains stable across influencer types. Consequently, for Gen Z and Millennials, virtual influencers function as sophisticated "interactive displays," sufficiently credible to drive a

transaction even in the absence of deep interpersonal trust. This indicates that the lack of a "soul" or "lived experience" creates a barrier to social bonding but not necessarily to the perceived financial risk regarding the product itself. Ultimately, we are entering an era where "authenticity" is bifurcated into *emotional authenticity* (human) and *functional authenticity* (virtual), proving that influence does not always require a heartbeat.

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Appendix

Survey Questionnaire

[Demographics]

- Please select your age group.
  - Below 18
  - 18 – 28
  - 29 – 44
  - Above 44
- Which of the following best describes your current occupation status?
  - Healthcare (e.g., nurse, doctor, medical assistant)
  - Technology (e.g., software developer, IT support)
  - Business (e.g., marketing, finance, management)
  - Student (e.g., college or graduate student)
  - Unemployed (currently not working)
  - Others (please specify below)
- What is the highest level of education you have completed?
  - Some high school
  - High school diploma or GED
  - Some college
  - Associate's degree or technical diploma
  - Bachelor's degree
  - Master's degree
  - Doctoral or professional degree (Ph.D., M.D., J.D., etc.)

[Social Behaviors]

- How frequently do you use social media platforms such as Twitter, Instagram, TikTok, Facebook, etc.?
  - Every day
  - Several times a week
  - About once a week
  - Less than once a week
  - Never

Condition A: Virtual Influencer

[Survey Introduction]

You will now view a social media post created by a virtual influencer.

"Virtual influencers are computer-generated characters with social media profiles who act like real-life influencers.

They often collaborate with brands, post lifestyle content, and engage with followers just like human influencers."

After viewing the post, you will be asked to answer a few questions based on your impressions.

<Showing the randomly assigned post>
Post for AI influencer: https://www.instagram.com/fit_aitana/reel/DO3tj63iHly/?hl=en

Participants will be shown an actual Instagram-style post sourced from an existing virtual influencer. The post features a skincare product, and is designed to match the human influencer version in terms of caption style, visual layout, and product placement. This ensures that the only difference across conditions is the type of influencer.

<After seeing the post>

[Manipulation check]

- Keep in mind that Aitana is a virtual influencer while answering the following questions. Virtual influencers are computer-generated characters with social media profiles who act like real-life influencers.
  Please confirm that you understand.
  - ☐ Yes
  - ☐ No
- What was the product featured in the advertising you just saw?
  - Sunglasses
  - Clothes
  - Skin Care
  - Sneakers

[Survey Questions]

Based on the post you viewed, please indicate your agreement with the following statements.

After viewing the post, please indicate your agreement with the following statements using a 5-point scale (1 = Strongly disagree, 5 = Strongly agree):

[Evaluation #1 – Product Recommendation]

Goal: Measure trust in the product recommendation itself — i.e., whether participants believe the product is good or worth considering because the influencer recommended it.

- I would not hesitate to take into account the product suggestions I can find in the Instagram posts published by this influencer.
- I would feel secure in following the suggestions about the product shown in this Instagram post.
- I would rely on the product recommendations made by this influencer.

[Evaluation #2 – Purchase Intention]

Goal: Capture behavioral intention — the likelihood that participants will act on the influencer's recommendation (buying, saving, clicking, or otherwise engaging with the product)

- I intend to buy the advertised product.
- I am likely to purchase the advertised product.
- I have high purchase interest in the advertised product.
- I will never buy the advertised product.
- I will probably buy the advertised product.

[Evaluation #3 – Trust and authenticity]

Goal: Assess source credibility and perceived authenticity of the influencer — how honest, credible, and genuine the influencer appears.

- Aitana is genuine.
- Aitana seems real to me.
- Aitana is authentic.
- I am confident in Aitana.
- Aitana provides security.
- Aitana has integrity.
- Aitana is reliable.
- I can trust Aitana.

[Evaluation #4 – Positive Feeling]

Goal: Measure affective response or liking of the influencer — how much participants personally like or feel favorably toward the influencer.

- Please rate your overall attitude towards this influencer.
  bad/good, favourable/unfavourable, like/dislike, negative/positive

<Shown after the main survey questions>

[Familiarity with Influencers]

You've completed the initial set of questions. Next, we'd like to ask a few more about your familiarity with virtual influencers.

"Virtual influencers are computer-generated characters with social media profiles who act like real-life influencers. They often collaborate with brands, post lifestyle content, and engage with followers just like human influencers."

Please answer the following questions:

- How familiar are you with virtual influencers?
  - (1 = Not at all familiar, 5 = Extremely familiar)
- Have you ever followed at least one virtual influencer on social media?
  - ☐ Yes
  - ☐ No

Scale Point Description

1 Strongly disagree

2 Somewhat disagree

3 Neither agree nor disagree

4 Somewhat agree

5 Strongly agree

Condition B: Human Influencer

Same questions as above only changing the name of the influencer, except for the following section.

[Survey Introduction]

You will now view a social media post created by a human influencer. After viewing the post, you will be asked to answer a few questions based on your impressions.